



Xi'an Jiaotong-Liverpool University

西交利物浦大學

RBE211TC DYNAMIC SYSTEMS

INTRODUCTION

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Educational Aims

1. Develop an understanding of modelling methods for mechanical systems.
2. Introduce the fundamental principles governing the behaviour of first- and second-order dynamic systems in the time domain.
3. Develop practical skills in designing mechatronic systems through conducting and reporting simple experiments.



Learning Outcomes

- A. Model dynamic systems and analyse their characteristics across mechanical, thermal, fluid, and electrical domains, demonstrating a comprehensive understanding.
- B. Apply essential principles governing the behavior of first and second order systems in the time domain, showcasing proficiency in practical applications.
- C. Apply advanced analytical methods to determine natural frequencies and mode shapes and understand the stability issues in mechanical systems.
- D. Design and optimise simple mechatronic systems to achieve the specified objectives related to performance, cost, safety, environmental protection and sustainability.



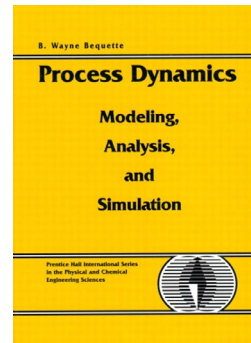
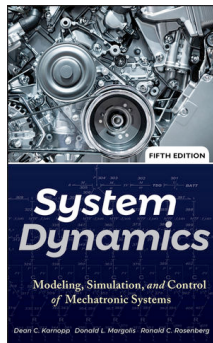
Assessment – LO Mapping

Learning Outcome	Laboratory (45%)	Final Exam (55%)
LO1 - Model dynamic systems and analyse their characteristics across mechanical, thermal, fluid, and electrical domains, demonstrating a comprehensive understanding.	✓	✓
LO2 - Apply essential principles governing the behavior of first and second order systems in the time domain, showcasing proficiency in practical applications.	✓	✓
LO3 - Apply advanced analytical methods to determine natural frequencies and mode shapes and understand the stability issues in mechanical systems.	✓	✓
LO4 - Design and optimise simple mechatronic systems to achieve the specified objectives related to performance, cost, safety, environmental protection and sustainability.	✓	

Recommended Readings

Reference Textbooks

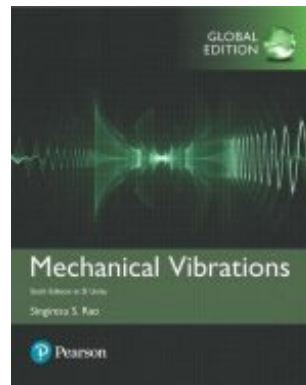
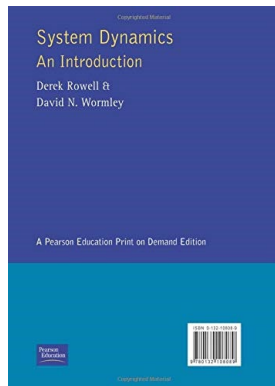
Title	Author	ISBN/Publisher
1 System Dynamics Modelling, Simulation, And Control of Mechatronic Systems, 5 th Edition	Dean C. Karnopp, Donald L. Margolis, Ronald C. Rosenberg	978-0470889084 /Wiley (2012)
2 Process Dynamics: Modelling, Analysis and Simulation	B. Wayne Bequette	0-13-206889-3 /Prentice Hall International (1998)



Recommended Readings

Reference Textbooks

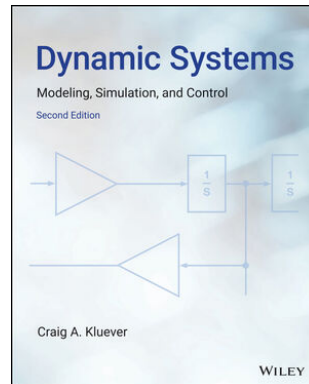
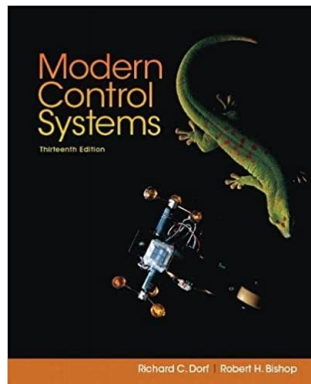
Title		Author	ISBN/Publisher
3	System Dynamics: An Introduction, 1 st Edition	Derek Rowell, David N. Warmley	978-0132108089 /Pearson (1996)
4	Mechanical Vibrations, 5 th Edition	Singiresu S. Rao	978-1-292-17860-8 /Pearson (2017)



Recommended Readings

Reference Textbooks

Title	Author	ISBN/Publisher
5 Modern Control Systems, 13 th Edition	Dorf & Bishop	978-0134407623 /Pearson (2016)
6 Dynamic Systems: Modeling, Simulation and Control, 2 nd Edition	Craig A. Kluever	9781119723479 /Wiley (2020)



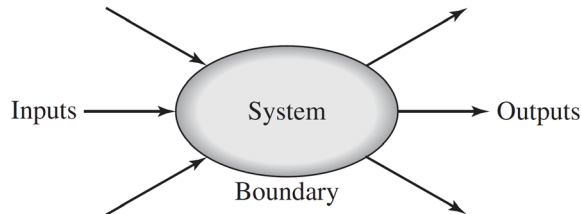
Lecture 1

INTRODUCTION TO DYNAMIC SYSTEMS



What is a System?

- **System:** A collection of interacting components that work together to perform a specified function or objective.
- **Components:** The parts that interact within the system (e.g., mechanical, electrical, biological).
- **Subsystem:** A smaller system within a larger system.
- **Environment:** The external conditions or influences that affect the system.

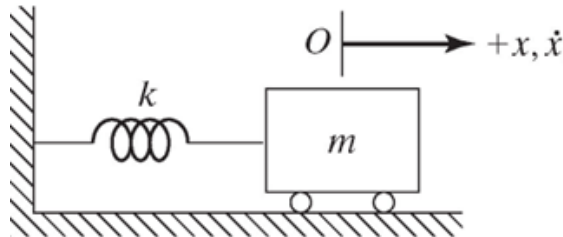


A system input-output diagram, showing the system boundary.



What is a Dynamic System?

- Dynamic System: A system in which the current output depends on past inputs and initial conditions.
- State Variables: The quantities that describe the system at any given time (e.g., position, velocity, pressure).
- Examples:
 - A mass-spring system with position and velocity as state variables.
 - A robotic arm with joint angles and joint velocities as state variables.



A mass-spring system.



Classification of Dynamic Systems

- Dynamic systems can be classified based on how their behaviour is represented mathematically.
- In general, engineering dynamic systems are classified into:
 - Lumped vs Distributed Systems
 - Continuous-Time vs Discrete-Time Systems
 - Time-Invariant vs Time-Varying Systems
 - Linear vs Nonlinear Systems



Classification of Dynamic Systems

Lumped vs Distributed Systems

- A distributed system has variables that vary continuously in space and time.
 - These systems are typically described using partial differential equations (PDEs).
- A lumped system assumes that spatial variations are negligible.
 - The system can therefore be described using a finite number of variables.
- Lumped systems are typically modelled using ordinary differential equations (ODEs).
- In this module, most physical systems will be treated as lumped systems.



Classification of Dynamic Systems

Continuous-Time vs Discrete-Time Systems

- A continuous-time system involves variables that are defined for all time.
 - These variables are typically functions of time, such as $x(t)$.
- A discrete-time system involves variables that are defined only at specific time instants.
 - These variables are typically written as $x[k]$ or $x(kT_s)$.
- Continuous-time systems are usually described by differential equations.
- Discrete-time systems are usually described by difference equations.
- In this module, most dynamic systems will be treated as continuous-time systems.



Classification of Dynamic Systems

Time-Varying vs Time-Invariant Systems

- A time-varying system has parameters that change with time.
 - The system behaviour may vary even if the input remains the same.
- A time-invariant system has parameters that remain constant over time.
 - The system response depends only on the input and initial conditions.
- Example of a time-varying system:
 - A mechanical system with changing mass or damping due to wear.
- Example of a time-invariant system:
 - A mass–spring–damper system with constant parameters.
- In this module, most dynamic systems will be treated as time-invariant systems.



Classification of Dynamic Systems

Linear vs Nonlinear Systems

- A linear system satisfies the principle of superposition:

1. Scaling:

$$u(t) \rightarrow y(t) \Rightarrow a u(t) \rightarrow a y(t)$$

2. Additivity:

$$u_1(t) \rightarrow y_1(t), \quad u_2(t) \rightarrow y_2(t)$$

$$u_1(t) + u_2(t) \rightarrow y_1(t) + y_2(t)$$

- A nonlinear system does not satisfy the superposition principle.
- Most real physical systems are inherently nonlinear.
- In practice, nonlinear systems are often linearised around an operating point.
- In this module, many systems will be approximated as linear systems for analysis.



Why Do We Need Mathematical Models?

- The behaviour of dynamic systems cannot be predicted without a mathematical representation.
- A mathematical model describes the relationship between system inputs and outputs.
- Models are developed by applying fundamental physical laws such as Newton's laws and Kirchhoff's laws.
- Mathematical models are typically expressed as differential equations.
- These models allow us to:
 - Analyse system behaviour
 - Predict system response
 - Design and optimise engineering systems



Module Roadmap

- Foundation
 - Introduction to Dynamic Systems
- System Modelling
 - Modelling of Mechanical Systems
 - Modelling of Electrical & Electromechanical Systems
 - Lagrangian Method
- System Analysis
 - Analytical Solution of Linear Systems
 - Vibration of Mechanical Systems
- System Control
 - Dynamic Systems in Feedback Control



Review

1. Module Introduction
2. What is a System?
3. What is a Dynamic System?
4. Classification of Dynamic Systems
5. Why Do We Need Mathematical Models?
6. Module Roadmap

